

When does wearable multi-signal fusion stop working?

A cross-session benchmark for fast biometric authentication

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Abstract

Consumer wearables can read several physiological signals at once: photoplethysmography (PPG), electrocardiography (ECG), electrodermal activity (GSR), and accelerometry (ACC). A recurring proposal is to fuse them into a fast "your body is your password" unlock: glance at the watch, get identified in seconds, like a fingerprint. Published fusion systems report near-perfect accuracy (equal error rates around 2%), which makes the idea look solved. We show that this number is an artefact of evaluating enrolment and test data from the **same recording session**. We build an open, data-source-agnostic benchmark that (i) under a leak-free evaluation reaches a comparably low within-session number on a public multi-signal dataset (Blasco et al. 2018; full-fusion EER 0.030, in the range of the reported ~ 0.02), and then (ii) re-evaluates the identical pipeline across sessions using two protocols: a cross-activity proxy on the same day, and a genuine cross-day protocol on a second dataset (PhysioNet Exam-Stress; three exams on different days). The within-session strength does not transfer: on Blasco the best honest cross-activity EER is 0.199 (about a 7x degradation), and on the true cross-day wrist dataset the signal is already weak within-session (EER 0.234) and stays weak across days (best 0.312). Signal fusion gives a small, real edge over the best single signal (+0.04 to +0.06 EER) but does not close the gap, and on both datasets a cardiac channel is dropped from the cross-session optimum because its within-session identity content does not transfer. Translated to a security operating point, the best honest cross-session fusion is $\sim 10,000$ to $16,000\times$ more permissive than Touch ID and $\sim 200,000$ to $312,000\times$ more permissive than Face ID. We conclude that fast wearable fusion is not a device-unlock primitive; its honest role is a low-stakes "probably still the same wearer" continuity check. All code, loaders, and a one-command reproduction are released.

1. Introduction

Every modern smartwatch carries an optical pulse sensor, an accelerometer, and increasingly an electrodermal or single-lead ECG sensor. A long line of work asks whether these signals can **identify** the wearer, and the most attractive framing is authentication: use the body's own signals as a password, so a device unlocks the moment it is worn. The literature appears to support it, with multi-signal fusion reporting equal error rates (EERs) around 2% or better.

There is a catch that is easy to miss and decisive in practice. Most flattering numbers split a **single recording session** into enrolment and test partitions. Within one session a classifier can lock onto session-specific nuisance structure (sensor placement, contact pressure, skin state) that is perfectly predictive *within that session* but has nothing to do with stable identity. A real unlock must work tomorrow, on a freshly donned device. The scientific question is therefore not "can we separate people within a session" (we can, easily) but "does that separation survive a change of session."

Contributions: (1) an open, data-source-agnostic benchmark (common schema + thin per-dataset loaders + one leakage-audited pipeline); (2) a competitive within-session number under a leak-free split (EER 0.030 on Blasco, in the range of ~ 0.02); (3) a cross-session evaluation under a same-day cross-activity proxy and a true cross-day protocol, showing a $\sim 7\times$ collapse on Blasco and a wrist signal that is weak within-session and stays weak across days; (4) a fusion verdict and security-bar accounting showing the honest operating point is 4-5 orders of magnitude short of Touch ID / Face ID. This is a negative result with a constructive framing: we measure precisely where a widely-assumed idea fails, and release the tooling so the measurement is cheap to repeat.

2. Related work

Physiological biometrics. PPG and ECG have both been studied as biometric traits, with within-session accuracies frequently in the high-90s percent. Fusion of cardiac signals with other modalities is a recurring theme.

The permanence problem. That intuition is rarely tested across time. Permanence, stability across sessions and days, is a named property in biometrics, and several wrist-PPG studies that test it report large cross-day degradation. The dataset we anchor on states its own limitation plainly: its signals "were acquired only once on a given day," and it recommends "cross-day and long-term analysis" as future work. This study is in part that recommended follow-up, extended to fusion and to a security-bar comparison.

Where this sits. We do not claim that fusing wearable signals to identify a person is new, nor that the within-session-inflation critique is new: a contemporaneous benchmark, ECG-biometrics-bench (arXiv:2605.01548), names the same "random-split" fallacy and evaluates it across seven ECG datasets with closed- and open-set protocols, and is more thorough on the single-signal cardiac case. Our contribution is the complementary *multi-signal fusion* slice it does not cover: whether fusing simultaneous signals, the mechanism most often proposed to reach a fast unlock, survives across sessions. We reach a competitive within-session number under a leak-free split with the same pipeline we then stress-test, report cross-session numbers with leakage controls and confidence intervals, and state the security implication in practitioner units.

3. Methods

3.1 Datasets

Dataset	Subjects	Sessions	Signals	Rate	Cross-session axis
Blasco 2018	25	3 activities (1 day)	PPG, ECG, GSR	100 Hz	cross-activity (proxy)
Exam-Stress	10	3 exams (different days)	PPG, ACC, GSR	64 Hz [†]	cross-day (true)

Blasco 2018. 25 subjects (ages 18-42, mean 28.1; 16 M, 9 F), three activity states on a single day (seated rest; walking; seated after a stroll). PPG/ECG/GSR at 100 Hz. There is **no usable accelerometer** in the released files, so Blasco fusion is PPG+ECG+GSR. Because all recordings are from one day, its cross-"session" axis is really **cross-activity**, a proxy.

PhysioNet Exam-Stress. Empatica E4 wrist device, 10 subjects, three exams on **different days**. BVP (PPG) 64 Hz, 3-axis ACC 32 Hz, EDA (GSR) 4 Hz; **no ECG**. Fusion is PPG+ACC+GSR; cross-

session axis is a genuine **cross-day** test. Recordings span 3 to 7 h. † Channels have different native rates; the loader resamples all onto a shared 64 Hz grid.

Together the two datasets cover all four signals and provide both a same-day proxy and a true cross-day protocol.

3.2 Common schema

A single `Recording` type holds {subject id, session id, condition, sampling rate, dict of equal-length time-aligned channels} with canonical channel names. Each dataset has a thin loader; no downstream code knows which dataset it is processing. This let us add a second dataset (different device, signals, rates) with **zero** change to the processing, feature, evaluation, or analysis code. `session_id` is the split axis.

3.3 Signal processing & windowing

Standard cleaning (NeuroKit2 for PPG/ECG; 5 Hz low-pass for GSR; ACC as vector magnitude). Non-overlapping **5-second windows**, the fast "glance-and-go" regime. Per-window quality flags mark bad channels; a window is kept if ≥ 1 cardiac channel is good, with per-channel "ok" flags for per-combination filtering.

3.4 Features

Signal-shape features only, never identity/recording tokens: cardiac HR/variability and pulse morphology, spectral band energy, distributional shape; GSR level/spread/slope; ACC level/spread/band energy. 26 features per dataset.

3.5 Leakage audit

Automated audit rejects identity-like names (token-boundary), confirms window index is never a feature, flags degenerate features, and red-flags implausible single-feature separability. Passes on both datasets. End-to-end control: within-session label permutation must score at chance, and it does (§4.5).

3.6 Evaluation

Closed-set verification; report EER and ROC AUC. Scorer = per-window RandomForest subject classifier (200 trees); verification score = $P(\text{claimed subject})$. We use a supervised classifier because that is the model class that produces the flattering literature numbers (a distance template gives within-session EER ≈ 0.17 and does not reproduce them). Standardisation fit on enrolment only. - **Within-session:** enrol/test on disjoint window partitions of the same session (runtime disjointness assertion), averaged over sessions. - **Cross-session:** enrol on one session, test on a different one (common subjects), averaged over ordered session pairs. Blasco = cross-activity; Exam-Stress = cross-day.

All randomness seeded; the whole pipeline regenerates via `reproduce.py`.

4. Results

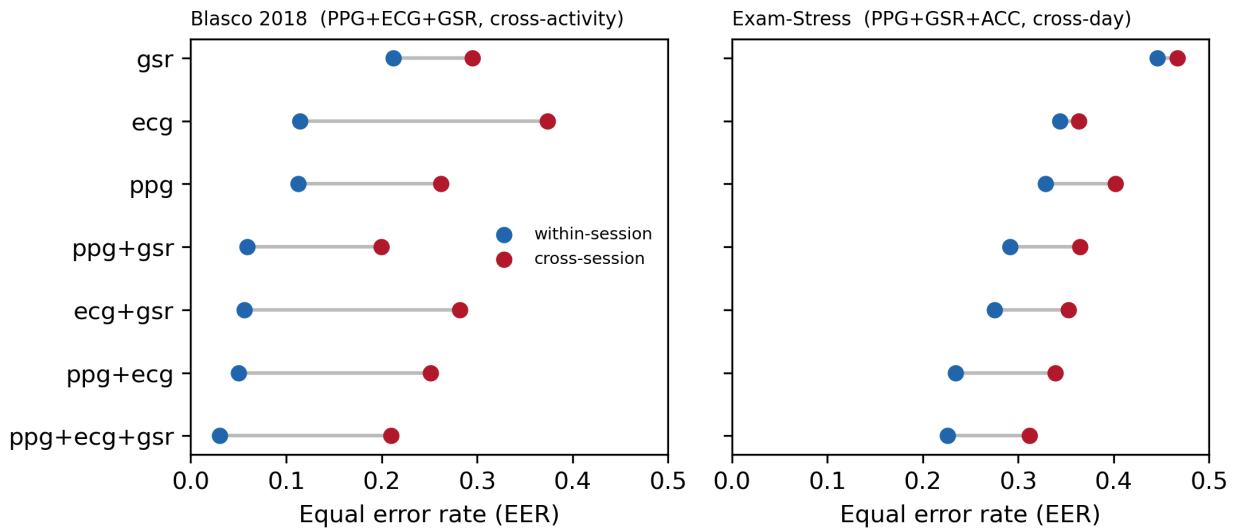
4.1 A strong within-session number, under a leak-free split

On Blasco, full-fusion within-session verification reaches **EER 0.030** under a leak-free time-blocked split, in the range of the ~ 0.02 reported by Blasco et al. This is a comparably low number, not a like-for-like reproduction (Blasco used a different scorer and 60 s training; we use a per-window RandomForest); it is enough to show our pipeline is competitive with the result that motivates the "body-as-password" premise, so the cross-session failure below is not a weak-model artefact. A prior version used a random-window split and reported EER 0.014; that split leaks temporally adjacent windows, and 0.030 (leak-free) is the honest number. Every single signal is individually usable within session (EER 0.113-0.212).

4.2 Accuracy collapses across sessions

Dataset	Signals	Within EER	Cross EER	Collapse
Blasco 2018	PPG+ECG+GSR	0.030	0.199-0.210	$\sim 6.9\times$ (same-combo)
Exam-Stress	PPG+GSR+ACC	0.234	0.312-0.339	$\sim 1.5\times$ (same-combo)

Within-session accuracy does not transfer across sessions



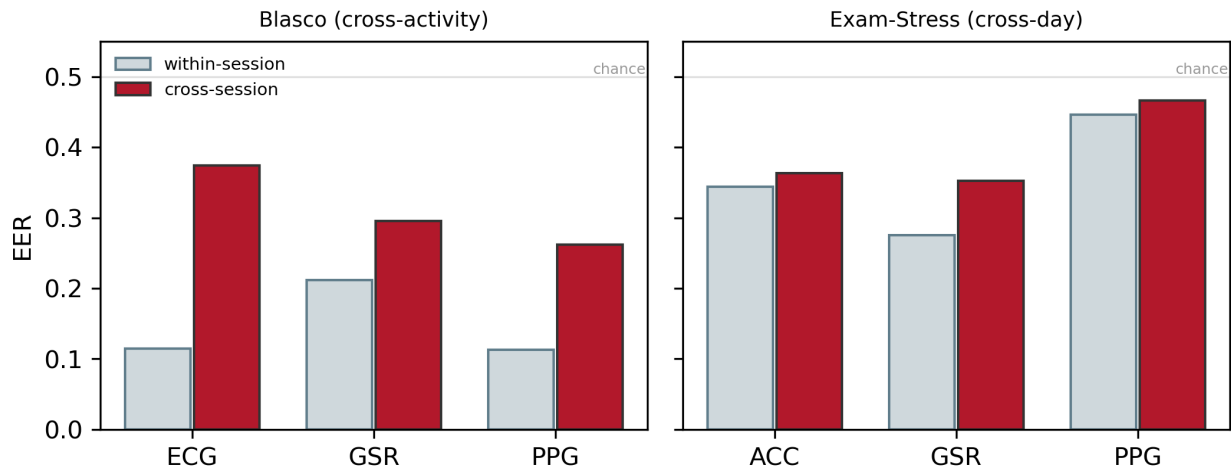
On Blasco the best cross-activity EER is 0.199 ($\sim 7\times$ same-combination degradation from the within-session 0.030). On Exam-Stress, the true cross-day test on a real wrist device, the signal is already weak within-session (full-fusion EER 0.234) and only degrades a further $\sim 1.5\times$ to 0.312 across days; there is no strong number to collapse from (a within-session baseline that is itself weaker because multi-hour wrist BVP is far noisier). At these operating points the system is close to a coin flip at any usable threshold.

4.3 Does fusion survive? A small edge, not a rescue

Fusion's cross-session advantage over the best single signal is real but small: +0.062 EER (Blasco), +0.041 (Exam-Stress). It does not close the gap; absolute cross-session EER stays 0.20-0.34. The full-fusion triple is never the cross-session optimum: the best combination is PPG+GSR on Blasco

(ECG drops out) and GSR+ACC on Exam-Stress (PPG drops out): in each dataset a **cardiac channel is discarded** because its within-session identity content does not transfer.

No single signal survives the move across sessions



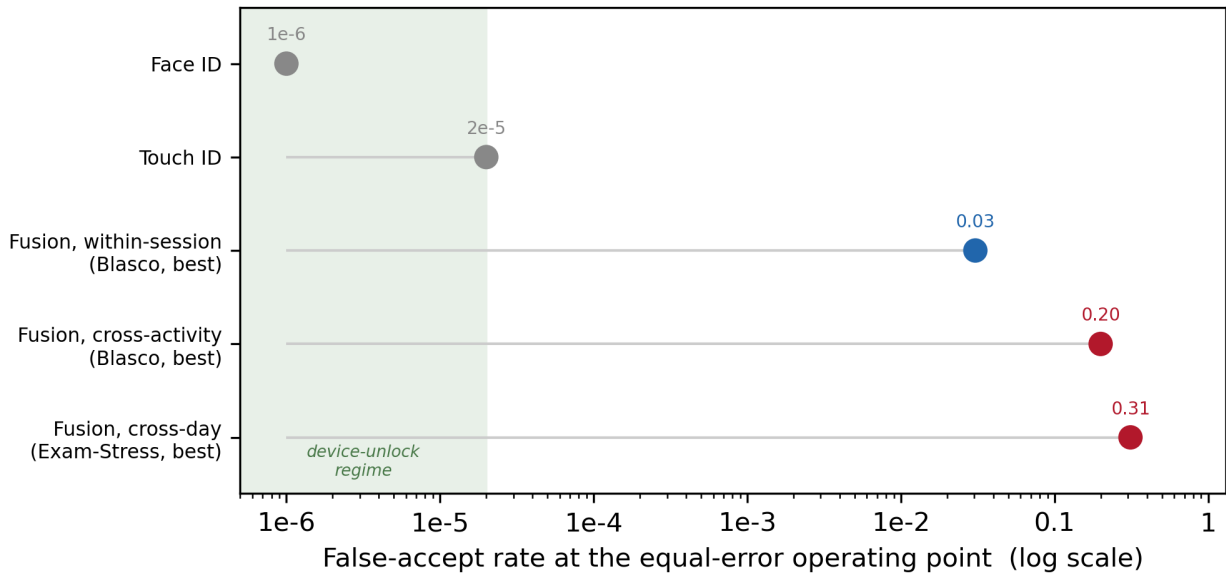
No single signal survives the move across sessions. On the wrist device (Exam-Stress) the cardiac PPG, the signal a consumer unlock would most naturally rely on, is the least useful single signal across days in absolute terms (cross-day EER 0.467, near chance) and is dropped from the optimum. Note there is no universal "cardiac degrades most" rule: on Blasco the cardiac ECG degrades most across sessions, while on Exam-Stress PPG degrades least (it barely worked within-session). The "fusion helps" edge is also a best-of-7-combinations selection on ~6 session-pair units (optimistically biased), and on Exam-Stress the full triple (0.339) is worse than 2-signal GSR+ACC (0.312), i.e. adding PPG hurts.

4.4 Security-bar accounting

At the equal-error operating point, false-accept rate = EER. Against Touch ID ($\text{FAR} \approx 2 \times 10^{-5}$, ~1 in 50,000) and Face ID ($\text{FAR} \approx 1 \times 10^{-6}$, ~1 in 1,000,000):

Operating point	Combination	FAR (=EER)	vs Touch ID	vs Face ID
Blasco cross-activity	PPG+GSR	0.199	~10,000x	~200,000x
Exam-Stress cross-day	GSR+ACC	0.312	~15,600x	~312,000x

Honest cross-session fusion is 4 to 5 orders of magnitude short of a real device unlock



The best honest cross-session fusion is 4-5 orders of magnitude short of a real device unlock. Two caveats keep this an order-of-magnitude, directional statement: the device FARs are manufacturer specs tuned to a low-FAR (high false-reject) point, whereas ours is the balanced equal-error point; and our EER is closed-set while Touch ID/Face ID are open-set, which would if anything widen the gap.

4.5 Robustness and honesty checks

- **Split-leakage check:** a random-window within-session split gives EER 0.014 (Blasco) / 0.147 (Exam-Stress); the leak-free time-blocked split gives 0.030 / 0.234. The difference is temporal-adjacency leakage; all within-session numbers here use the leak-free split.
- **Label-shuffle:** permuting subject labels within session drives both within and cross EER to chance (Blasco within 0.030->0.502, cross 0.210->0.497; Exam within 0.234->0.497, cross 0.339->0.497); real performance is genuine signal, not leakage.
- **Window length:** Blasco features at 3/5/10 s keep within EER 0.022-0.038 and cross ~0.21, so the cross-session failure is not a windowing artefact.
- **Bootstrap CIs:** within vs cross 95% CIs do not overlap (Blasco within [0.014, 0.047] vs cross [0.119, 0.297]; Exam-Stress within [0.226, 0.249] vs cross [0.295, 0.377]), but are resampled over few units (3 within-session sessions, 6 cross-session pairs), so they describe session-level not subject-level variation.

5. Discussion

What the number means. The within-session ~2% EER is real but answers the wrong question. A password lives in the cross-session regime, where the honest number is 0.20-0.34 EER: unusable as an unlock.

Why fusion does not save it. Fusion gives a small real gain, dwarfed by the cross-session failure; in each dataset a cardiac channel drops out of the cross-session optimum (ECG on Blasco, PPG on the

wrist device). There is no universal "cardiac degrades most" rule; the consistent finding is that no channel reaches a usable cross-session operating point. Fusion cannot rescue signals that individually fail to transfer.

The honest use case. A fast fused read cannot be a device unlock, but it can be a low-stakes **continuity** check ("is this probably still the same wearer") where a 20% error rate is tolerable because the fallback is benign. Presenting the same technology as a fingerprint replacement is the error we forestall.

Limitations. (1) Blasco's cross axis is cross-activity on one day, a proxy; the true cross-day result (Exam-Stress) is worse. (2) The two datasets' absolute EERs are not directly comparable; we compare the within→cross collapse within each. (3) Cross-day, the residual is carried by GSR/ACC, which may encode behavioural/postural habit or per-session stress baseline rather than stable identity; we do not claim the residual is "identity." (4) Both datasets are small ($n=25$, $n=10$); direction is robust to our controls, but magnitudes on larger populations remain to be measured, which the released benchmark enables.

6. Conclusion

Fast multi-signal wearable fusion is strong within a single session (Blasco full-fusion EER 0.030 under a leak-free split) and does not transfer: it degrades about 7x across activity states on the same day, and on a real wrist device across days it never authenticated well even within a session (EER 0.234) and stays weak (0.312). Either way the honest cross-session operating point is 4-5 orders of magnitude short of a real device unlock. Fusion helps a little and does not close the gap; a cardiac channel drops out of the cross-session optimum in each dataset. We release a reproducible, leakage-audited, cross-session-honest benchmark and a one-command reproduction.

Reproducibility & data availability

All code, loaders, leakage audit, evaluation harness, and `reproduce.py` are at <https://github.com/noorabdalla04/wearable-fusion-auth-benchmark>. Raw datasets come from their original sources under their own licenses (Blasco 2018, DOI 10.3390/s18092782; PhysioNet Exam-Stress, ODC-By); the repo never redistributes raw data and documents provenance and checksums. Results were produced with a fixed random seed.

References

- Blasco, J. & Peris-Lopez, P. (2018). *On the Feasibility of Low-Cost Wearable Sensors for Multi-Modal Biometric Verification*. Sensors 18(9), 2782. doi:10.3390/s18092782
- *A Wearable Exam Stress Dataset for Predicting Cognitive Performance in Real-World Settings*. PhysioNet v1.0.0 (ODC-By). <https://physionet.org/content/wearable-exam-stress/1.0.0/>
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